

The purpose of this worksheet is to develop familiarity with the Fourier transform by deriving a set of transforms that are listed in Table 5-6 of your textbook. Please derive the following forward and inverse Fourier transforms. The correct answers are in Table 5-6, so your goal should be to practice your ability to perform these derivations correctly.

1. (50 points) Calculate the Fourier transforms for the following signals:

$$x_1(t) = \delta(t - T)$$

$$x_2(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

$$x_3(t) = te^{-at}u(t)$$

$$x_4(t) = \cos(\omega_0 t)u(t)$$

$$x_5(t) = e^{-at} \cos(\omega_0 t)u(t)$$

2. (50 points) Calculate the inverse Fourier transforms for the following signals:

$$\hat{X}_1(\omega) = 1$$

$$\hat{X}_2(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$\hat{X}_3(\omega) = \frac{1}{a - j\omega}$$

$$\hat{X}_4(\omega) = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\hat{X}_5(\omega) = \frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$$

1. (50 points) Calculate the Fourier transforms for the following signals:

$$x_1(t) = \delta(t - T)$$

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$$x_5(t) = e^{-at} \cos(\omega_0 t)u(t)$$

$$X_1(t) = \delta(t - T)$$

$$\hat{X}(\omega) = \int_{-\infty}^{\infty} \delta(t - T) e^{-j\omega t} dt$$

$$\hat{X}(\omega) = e^{-j\omega T}$$

$$X_2(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

The rect function is 1 for $-\frac{\tau}{2} < t < \frac{\tau}{2}$ and 0 elsewhere

$$\text{Calculation } \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\tau/2}^{\tau/2} = \frac{e^{-j\omega \tau/2} - e^{j\omega \tau/2}}{-j\omega} \Rightarrow \text{using Euler's formula, this becomes } \frac{2 \sin(\frac{\omega \tau}{2})}{\omega} = \frac{\tau \sin(\frac{\omega \tau}{2})}{\frac{\omega \tau}{2}} = \tau \text{sinc}\left(\frac{\omega \tau}{2}\right)$$

$$X_2(\omega) = \tau \text{sinc}\left(\frac{\omega \tau}{2}\right)$$

$$X_3(t) = te^{-at}u(t)$$

$$X_3(\omega) = j \frac{d}{d\omega} X(\omega) : e^{-at} \Rightarrow \frac{1}{a + j\omega}$$

$$X_3(\omega) = j \frac{d}{d\omega} \left[\frac{1}{a + j\omega} \right] = j \left[-j(a + j\omega)^{-2} \right] = \frac{-j^2}{(a + j\omega)^2} = \frac{1}{(a + j\omega)^2}$$

$$X_4(t) = \cos(\omega_0 t)u(t)$$

$$\text{Using } \cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$X_4(t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) u(t)$$

$$\text{Using frequency shift property: } e^{j\omega_0 t} X(t) \Rightarrow X(\omega - \omega_0)$$

$$u(t) \Rightarrow \pi \delta(\omega) + \frac{1}{j\omega}$$

$$e^{j\omega_0 t} \Rightarrow X(\omega - \omega_0)$$

$$e^{-j\omega_0 t} \Rightarrow X(\omega + \omega_0)$$

$$\Rightarrow \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)] \left[\pi \delta(\omega) + \frac{1}{j\omega} \right]$$

$$\Rightarrow \frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$$

$$X_5(t) = e^{-at} \cos(\omega_0 t)u(t)$$

$$e^{-at} \Rightarrow \frac{1}{a + j\omega} : \text{frequency shift}$$

$$\text{Using } \cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$\frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$\text{shift} \Rightarrow \frac{1}{2} (e^{-(a-j\omega)t} + e^{-(a+j\omega)t})$$

$$\Rightarrow \text{Using frequency shift property: } e^{j\omega_0 t} X(t) \Rightarrow X(\omega - \omega_0)$$

$$\Rightarrow \frac{1}{2} \left(\frac{1}{a + j(\omega - \omega_0)} + \frac{1}{a + j(\omega + \omega_0)} \right)$$

2. (50 points) Calculate the inverse Fourier transforms for the following signals:

$$\hat{X}_1(\omega) = 1$$

$$\hat{X}_2(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$\hat{X}_3(\omega) = \frac{1}{a - j\omega}$$

$$\hat{X}_4(\omega) = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\hat{X}_5(\omega) = \frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$$

$$X_1(\omega) = 1$$

$$x_1(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 1 e^{j\omega t} d\omega \Rightarrow \delta(t) \text{ by duality}$$

$$x_2(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$= F_x^{-1}[\pi\delta(\omega)] = 1/2$$

$$= F_x^{-1}\left[\frac{1}{j\omega}\right] = 1/2 \operatorname{sgn}(t)$$

$$= 1/2 + 1/2 \operatorname{sgn}(t) \Rightarrow u(t)$$

$$x_2(t) = u(t)$$

$$X_3(\omega) = \frac{1}{a - j\omega}$$

$$= F^{-1}[e^{-at} u(t)] = \frac{1}{a + j\omega}$$

$$\text{time reversal} = \frac{1}{a - j\omega}$$

$$x_3(t) = \frac{1}{a - j\omega}$$

$$x_4(\omega) = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\text{using } F[\sin(\omega_0 t)] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$x_4(t) = \sin(\omega_0 t)$$

$$x_5(\omega) = \frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$$

$$F[e^{-at} \sin(\omega_0 t) u(t)] = \frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$$

$$x_5(t) = e^{-at} \sin(\omega_0 t) u(t)$$